

Total No. of Questions : 10]

SEAT No. :

P3735

[5561]-132

[Total No. of Pages : 3

B.E. (Mechanical)

DYNAMICS OF MACHINERY

(2012 Course) (402043) (End Semester) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

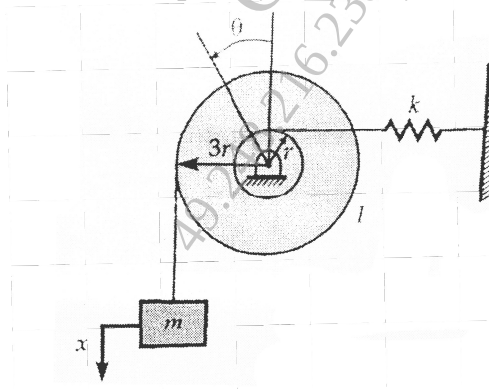
- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Draw neat diagrams wherever necessary.
- 3) Use of scientific calculator is allowed.
- 4) Assume suitable data wherever necessary.
- 5) Figures to the right indicate full marks.

Q1) A six cylinder inline reciprocating engine has a firing order 1-4-5-2-3-6. The mass of reciprocating part per cylinder is 3 kg. The length of cranks and connecting rods are 50 mm and 200 mm respectively. The cylinders are spaced at 300 mm pitch. If the engine runs at 1000 rpm Determine **[10]**

- a) Primary and secondary unbalance force if any.
- b) Primary unbalance couple with reference to central plane.

OR

Q2) a) Find natural frequency for the system given below. **[5]**



- b) Explain Coulomb's damping. And state two differences between Coulomb's damping and Viscous damping. **[5]**

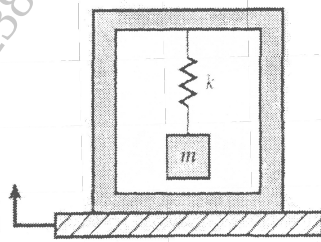
P.T.O.

- Q3) a)** Define following terms : **[4]**
- i) Magnification factor.
 - ii) Logarithmic Decrement.
 - iii) Damping Factor.
 - iv) Quality Factor.
- b) In a vibratory system vehicle is to be designed with following parameters
 $K = 100 \text{ N/m}$, $C = 2 \text{ Nsec/m}$, $m = 1 \text{ kg}$. **[6]**
 Calculate decrease in amplitude from its starting value after 3 complete oscillations and damped frequency of oscillation.
- OR
- Q4) a)** The static deflection of an automobile on its springs is 10 cm. Find the critical speed when the automobile is travelling on the road which is approximated by a sine curve of amplitude 8 cm and a wavelength of 16 m. Assume the damping to be given by 0.05. Also determine the amplitude of vibration at 75 km/hr. **[5]**
- b) A body of 5 kg is supported on a spring of stiffness 200 N/m and has dashpot connected to it which produces resistance of 0.002 N at a velocity of 1 cm/sec. In what ratio will be the amplitude of vibration be reduced after 5 cycles. **[5]**
- Q5) a)** What are 2 rotor system? Derive a relation to determine the natural frequencies of a two rotor system. **[10]**
- b) Explain with neat diagram mathematical model of a motorbike. **[6]**
- OR
- Q6) a)** Explain Principal Modes of vibration for two DOF rectilinear vibratory system having 2 equal masses "m" connected by three springs having equal stiffness "k". Draw suitable diagrams of vibratory systems and modes of vibrations. **[10]**
- b) Explain with neat diagram mathematical model of a bicycle. **[6]**
- Q7) a)** Draw and explain transmissibility versus frequency ratio curves. **[6]**
- b) Name any instrument to determine the natural frequency of badminton racket. Explain detail working of that instrument with block diagram. **[6]**

- c) An electric motor weighs 25 kg and is mounted on rubber pad which deflects by 1 mm to motor weight. The rotor weighs 5 kg, has an eccentricity of 0.1 mm and rotates at 1500 rpm. Find the force transmitted to the foundation under the following conditions : [6]
- There is no damping.
 - Damping factor is 0.1.

OR

- Q8)** a) What is vibration absorber? Name different types of absorbers used and explain any one in detail. [6]
- b) A 3 kg mass is suspended in a box by a spring as shown in following fig. The box is put on a platform having vibration $y = 0.8 \sin 6t$ cm. Determine the absolute amplitude of the mass. Take $k = 6000$ N/m. [6]



- c) Explain following vibration Exciters : [6]
- Mechanical Exciter.
 - Electrodynamics Exciter.

- Q9)** a) Explain the working of sound meter with neat diagram. [6]
- b) Determine the sound pressure level for a sound with rms sound pressure of 2 N/m^2 and 0.4 N/m^2 . [4]
- c) Explain Reverberation chamber and Anechoic Chamber. [6]

OR

- Q10)** a) What is sound enclosure? Describe any one type of sound enclosure? [6]
- b) Define following terms : [4]
- Sound power level.
 - Sound pressure level.
 - Sound intensity.
 - Decibel scale.
- c) An operator in textile mill is operating five machines. The sound pressure levels of the machines at his position are 95 dB, 90 dB, 92 dB, 88 dB and 82 dB respectively. What is the total SPL at his position due to all the five machines? What is the SPL if the Machine 1 is turned off? Also find the SPL if machine 2 and 3 is turned off. [6]

